Prominence and Directed Consumer Search: Theory and Experiment

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Paid Prominence Directs Consumer Search

In various markets, some products are displayed more **prominent** than others, e.g. Google Ads, Amazon Sponsored Products, geographic locations of firms.

Paid Prominence: Firms compete for attention and pay to stand out.

Consumer Search: Costly inspection of each alternative.

Directed Search: Pre-search product characteristics (e.g. prominence, price) determine consumers' search decisions. An optimal stopping problem.

This paper: When firms' valuations of prominence are *endogenous*, how does paid prominence shape equilibrium prices, consumer behavior, and efficiency?

How Paid Prominence Affects Market Outcomes?

1. **Bidding:** How much do firms pay for prominence, and is bidding competitive?
2. **Pricing:** Does the prominent firm price higher or lower than the non-prominent firm?
3. **Search:** Does consumer search respond to prices and prominence?
4. **Welfare:** How does greater search prominence affect total surplus and its distribution?

Model: Three-Stage Game

Stage I
Auction

Two firms bid for the **top position**

Second-price auction

Stage II
Pricing

Both firms post **transparent** prices simultaneously p_1, p_2

Stage III
Search & Purchase

Buyer sees p_1, p_2; learns V_1 **for free**

May pay s to inspect Firm 2

Buys or exits

Model: Payoffs and Parameters

Buyer utility from buying at slot $j \in \{1, 2\}$:

$$u_j \;=\; V_j - p_j - s \cdot \mathbf{1}_{\{j=2\}}$$

Match value $V_j \in \{V, 0\}$ i.i.d., $\Pr(V_j{=}V) = q$, revealed upon visit. Free recall. Outside option $u_0 = 0$.

Seller profit:

$$\pi_j \;=\; p_j \cdot \mathbf{1}_{\{\text{sale}\}} \;-\; \text{position payment}_j$$

- **Top seller** (auction winner): position payment $= b^{(2)}$, the losing bid
- **Second seller** (auction loser): position payment $= 0$

Parameters: $V = 100, \quad q = 0.5, \quad s \in \{20, 40\}$

Consumer Search Condition

After observing (p_1, p_2) and inspecting V_1 for free, the consumer would search Firm 2 iff:

Under $V_1 = V$ (good match with Firm 1)

$$p_1 - p_2 \;\geq\; \frac{s}{q}$$

i.e. the price gap is large enough relative to the search cost.

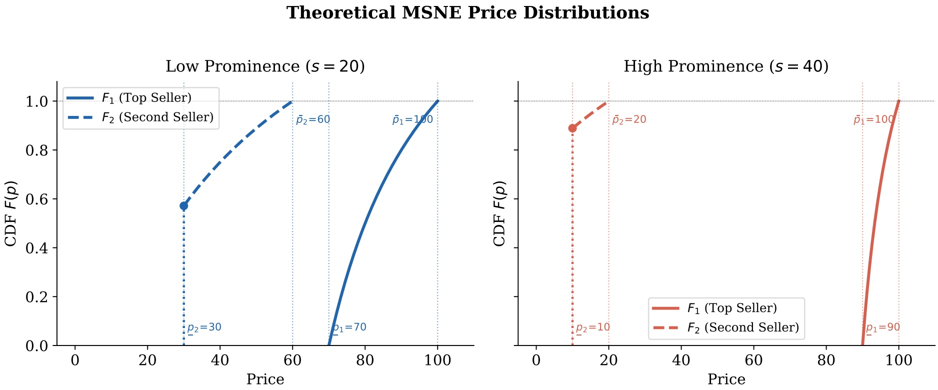
Under $V_1 = 0$ (bad match with Firm 1)

$$p_2 \;\leq\; V - \frac{s}{q}$$

i.e. net option value of inspecting Firm 2 is weakly positive.

Because of discontinuity and flatness in demand functions, there is no pure-strategy equilibrium.

Equilibrium: Mixed-Strategy Pricing CDFs



Both firms mix over prices. Firm 2 places an atom at its lower bound $\underline{p}_2$ — a **fire-sale** commitment to attract search. As s rises, supports separate and the gap widens.

Key Prediction: Prominence Confers Market Power

The prominent firm's price CDF **first-order stochastically dominates** the non-prominent firm's:

$$F_1(p) \;\leq\; F_2(p) \quad \text{for all } p$$

As s increases, both bounds of F_1 shift up and both bounds of F_2 shift down — the price gap widens.

Bidding for Prominence

Each firm bids its incremental value of the top position:

$$b^* \;=\; \bar{\Pi}_1 - \bar{\Pi}_2 \;=\; (1-q)\,s$$

With $q = 0.5$: $\quad b^*(s{=}20) = 10, \quad b^*(s{=}40) = 20.$

Net profits are equal and positive across positions:

$$\Pi^{\text{net}} \;=\; \bar{\Pi}_2 \;=\; \left(V - \frac{s}{q}\right) q(1-q)$$

The auction dissipates only the *incremental* value of prominence, not all rents.

Experimental Design

Treatment variation: search cost s (= visit cost for Second Seller)

- **Low Prominence:** $s = 20$
- **High Prominence:** $s = 40$

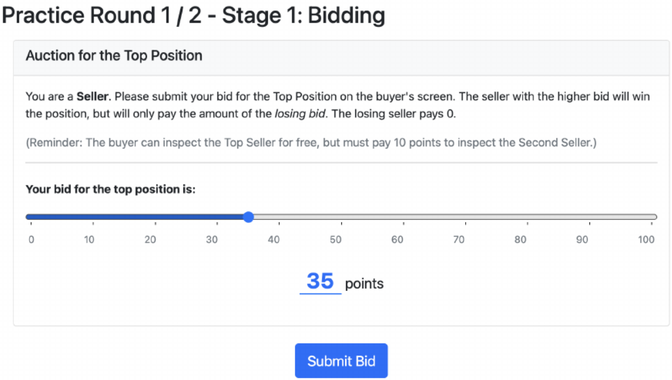
Fixed: $V = 100$, $q = 0.5$, 30 formal rounds (+ 2 practice)

Sessions: 3 per treatment, 15 subjects each. Between-subject.

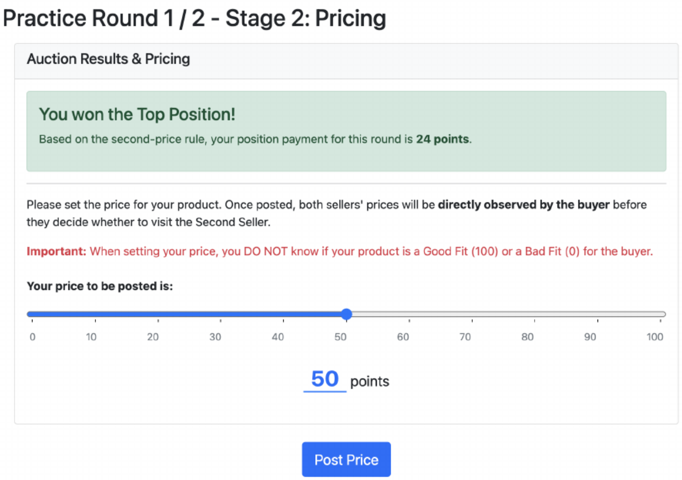
Total: 870 group-round observations (420 at $s = 20$; 450 at $s = 40$)

UVA-IRB-SBS 8298. Show-up \$10 + one random round (10 pts = \$1).

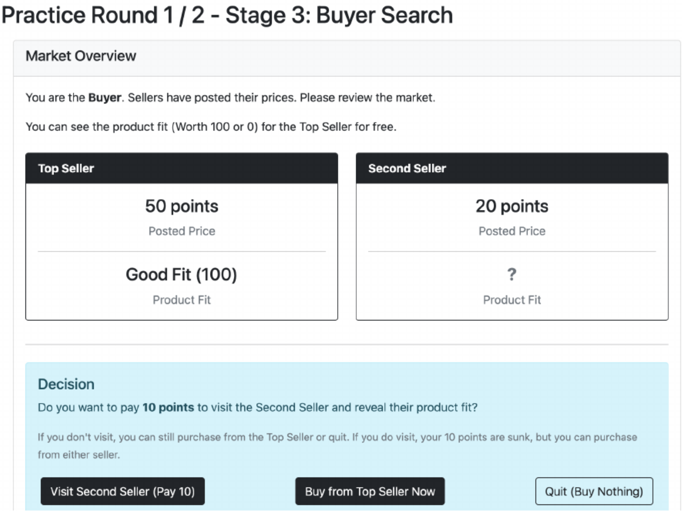
Interface: Seller Submits Bid



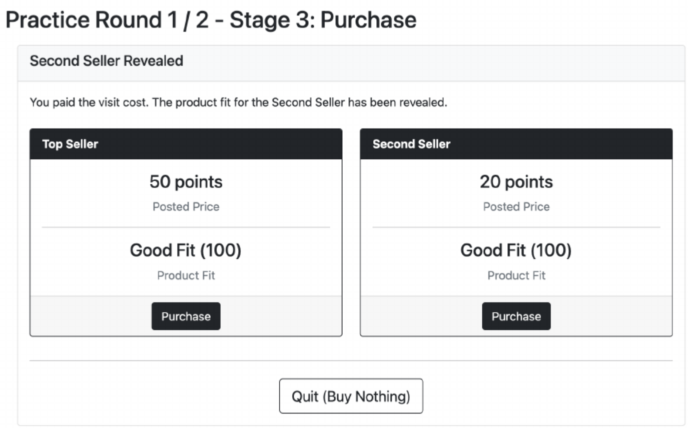
Interface: Seller Posts Price



Interface: Buyer Searches



Interface: Buyer Purchases



Interface: Feedback

Practice Round 1 / 2 - Round Results

Round Summary

Market Outcomes

Position	Price Posted	Product Fit
Top Seller	50 pts	Good Fit (100)
Second Seller	20 pts	Good Fit (100)

Buyer's Decision

- **Visited Second Seller?** Yes (Paid 10 pts)
- **Purchased From:** Second Seller

Your Points This Round

Payoff Points Earned: **-24 points**

starting (100) + earned (-24 points) = **76 points**

Reminder: This was a practice round and does not affect your final payment.

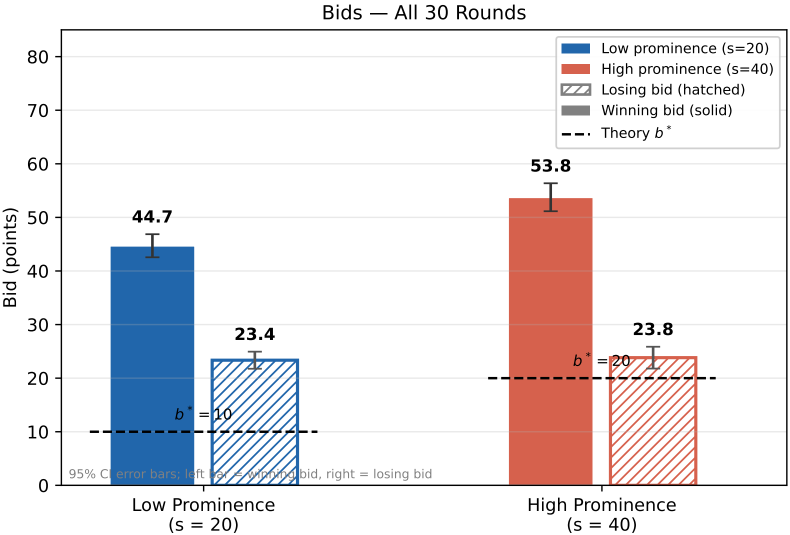
Theory Predictions by Treatment

Quantity	Low ($s = 20$)	High ($s = 40$)
Equilibrium bid b^*	10	20
Top seller support $[\underline{p}_1, \bar{p}_1]$	[70, 100]	[90, 100]
2nd seller support $[\underline{p}_2, \bar{p}_2]$	[30, 60]	[10, 20]
Fire-sale mass m_2	0.57	0.89
Search cutoff ($V_1{=}V$): s/q	40	80
Search cutoff ($V_1{=}0$): $V - s/q$	60	20
Net profit per firm	15	5

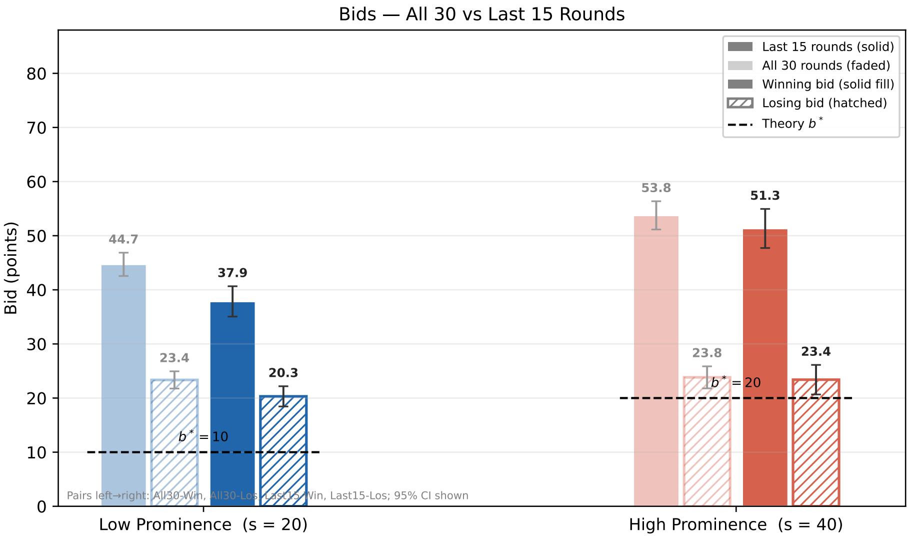
Data Overview

	$s = 20$	$s = 40$
Group-round observations	420	450
Avg. winning bid	44.7	53.8
Avg. top seller price	61.8	66.1
Avg. second seller price	48.9	39.1
Avg. price gap $p_1 - p_2$	12.9	27.0
Search rate (overall)	40%	25%
when $V_1 = V$	13%	10%
when $V_1 = 0$	66%	42%
Exit rate (unconditional)	33%	40%
when $V_1 = V$	0%	3%
when $V_1 = 0$	64%	80%

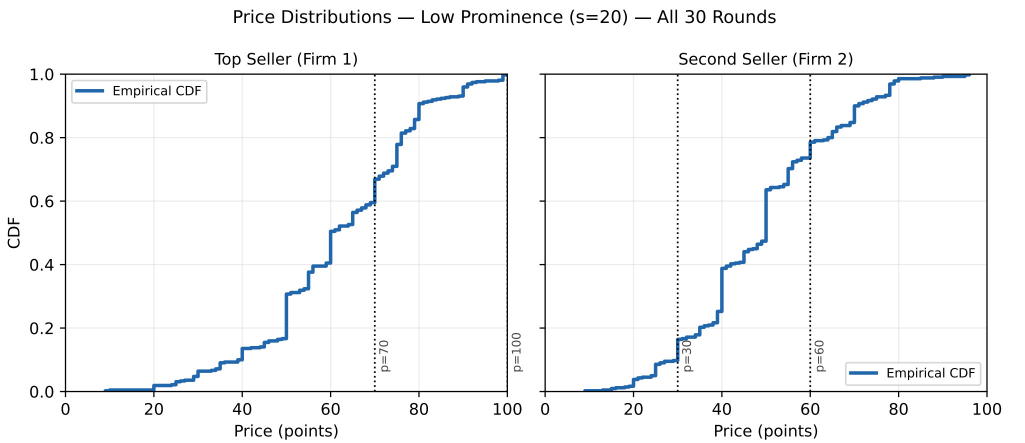
Bids — All 30 Rounds



Bids — Last 15 Rounds

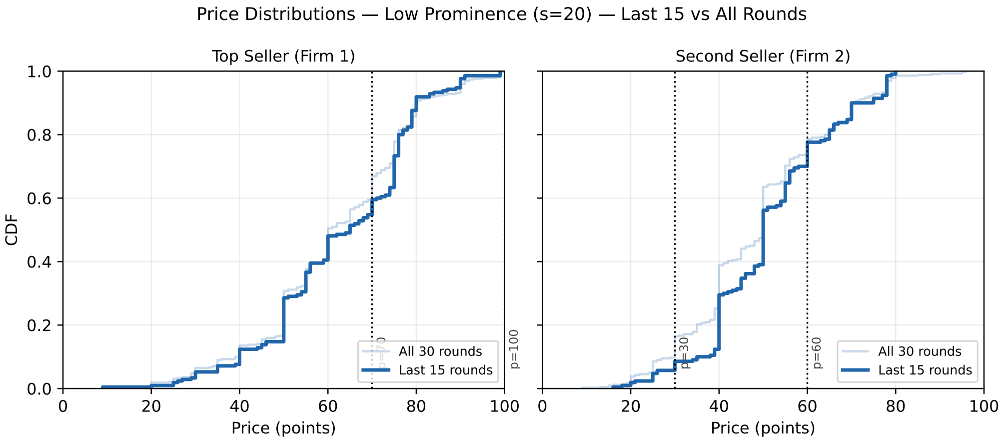


Price Distributions ($s = 20$) — All 30 Rounds



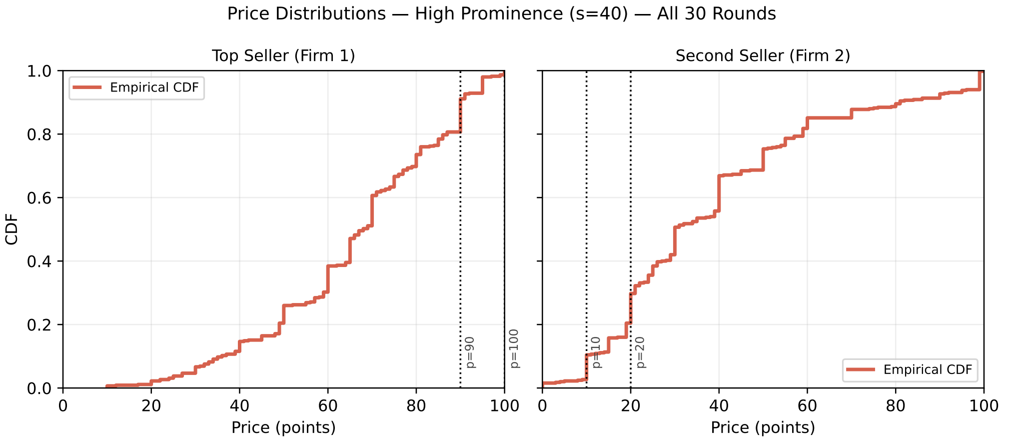
Price ordering confirmed: top seller's CDF lies to the right of second seller's. Distributions are more diffuse than the tight theoretical supports, but the qualitative prediction holds.

Price Distributions ($s = 20$) — Last 15 Rounds



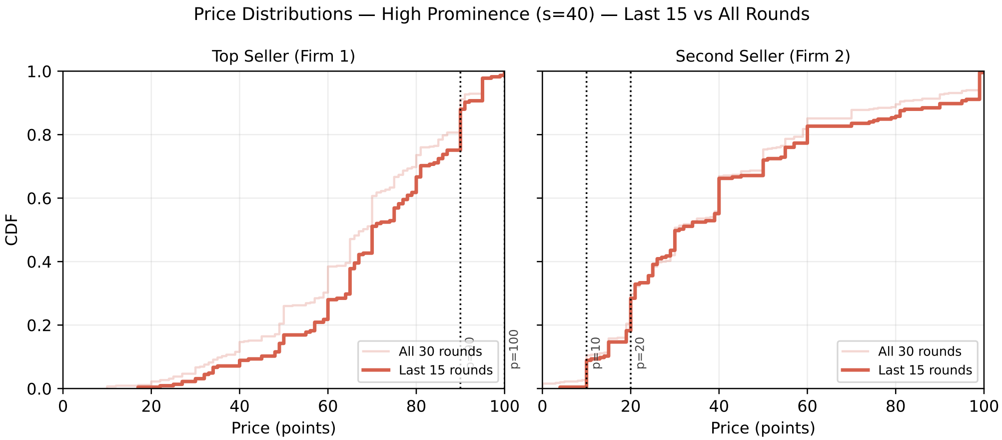
Both prices drift upward in later rounds (2nd seller: 46.2 $\rightarrow$ 51.7, ***). Sellers learn that buyers search less, reducing competitive pressure. The price ordering is maintained.

Price Distributions ($s = 40$) — All 30 Rounds



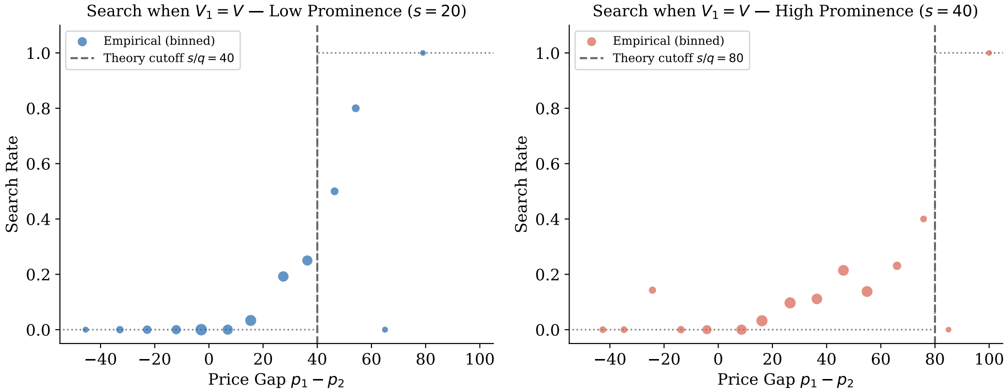
Theory predicts a very compressed second-seller support $[10, 20]$. Subjects price far above these bounds, but the ordering (Firm 1 prices higher) holds throughout.

Price Distributions ($s = 40$) — Last 15 Rounds



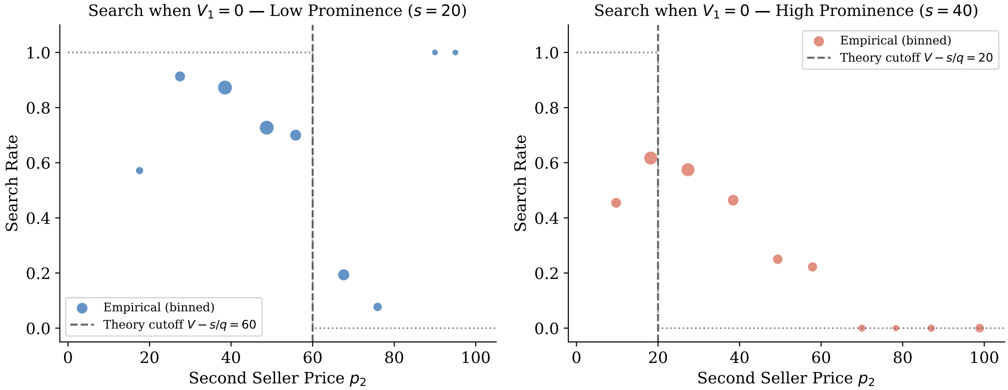
Top seller prices rise sharply (61.4 → 70.8, ***). Top sellers learn to exploit their lock-in advantage more aggressively as the experiment progresses.

Search Behavior when $V_1 = V$ — All 30 Rounds



Theory: search iff $p_1 - p_2 \geq s/q$ (40 or 80). Observed: smooth transition rather than a sharp step. Buyers are too willing to search even at small price gaps.

Search Behavior when $V_1 = 0$ — All 30 Rounds



Theory: search iff $p_2 \leq V - s/q$ (60 or 20). Under $s = 20$ most buyers comply. Under $s = 40$ the cutoff is very stringent (20) and many buyers search above it.

Measuring Social Surplus

Prices are transfers; total surplus depends only on *what* is allocated and *whether* search occurs:

$$SS^{\text{realized}} \;=\; V_a \cdot \mathbf{1}_{\{a\neq 0\}} \;-\; s \cdot \mathbf{1}_{\{\text{search}\}}$$

Decomposes into four pieces:

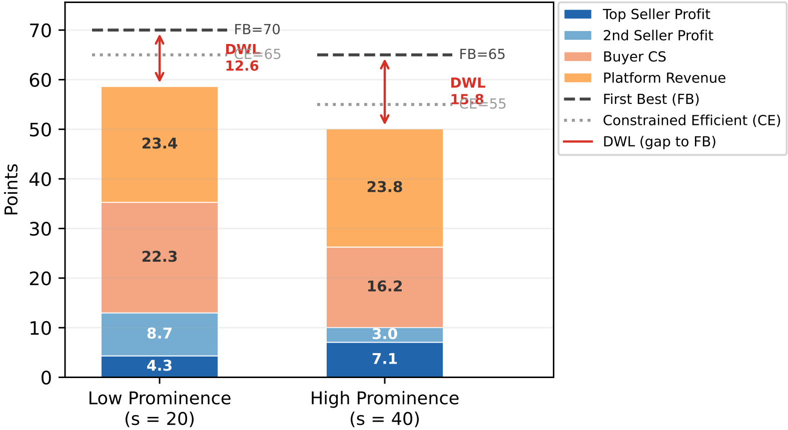
$$SS^{\text{realized}} = \pi_1^{\text{top}} + \pi_2^{\text{2nd}} + CS^{\text{buyer}} + \pi^{\text{plat}}$$

where $\pi^{\text{plat}} = b^{(2)}$, and all prices cancel as transfers.

Two benchmarks:

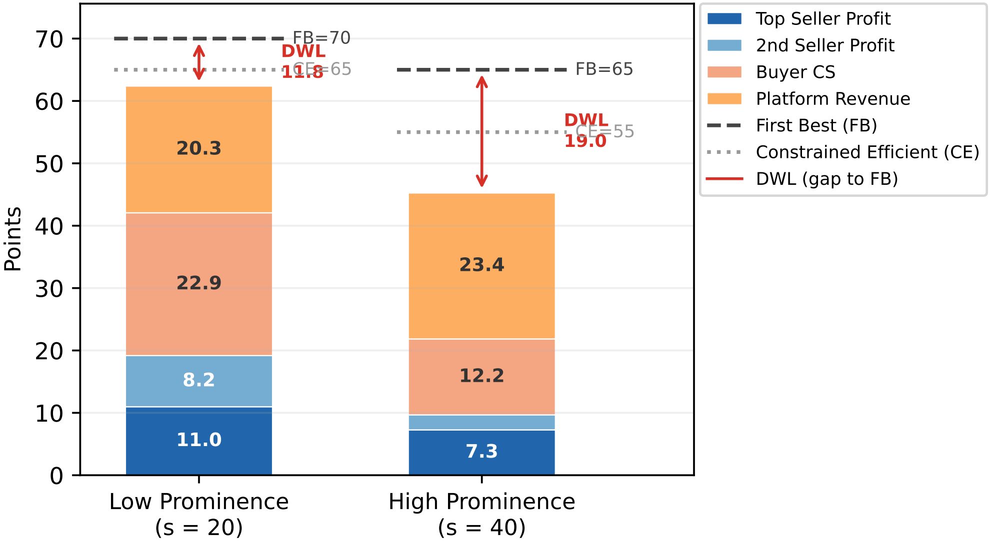
- **First Best (FB)**: planner observes both V_1 and V_2. $\mathbb{E}[SS^{FB}] = 75 - 0.25s$
- **Constrained Efficient (CE)**: planner faces same info as buyer. $\mathbb{E}[SS^{CE}] = 75 - 0.5s$

Welfare Decomposition — All 30 Rounds

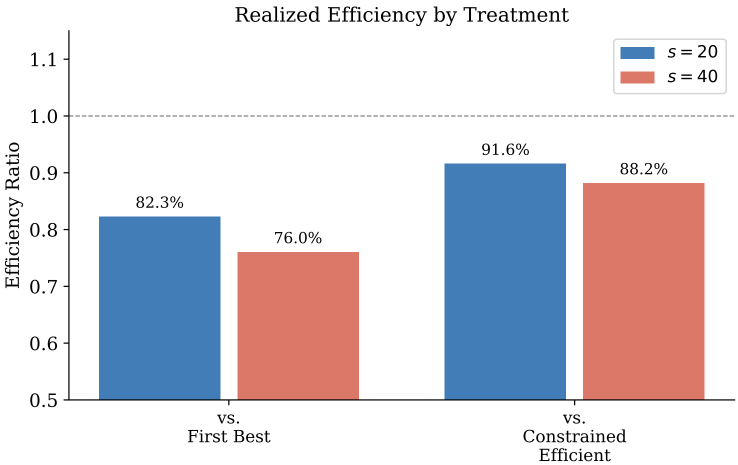


Higher prominence shifts surplus from buyers and second seller toward the top seller. Platform revenue is nearly flat. DWL (red) rises from 12.6 to 15.8 points.

Welfare Decomposition — Last 15 Rounds

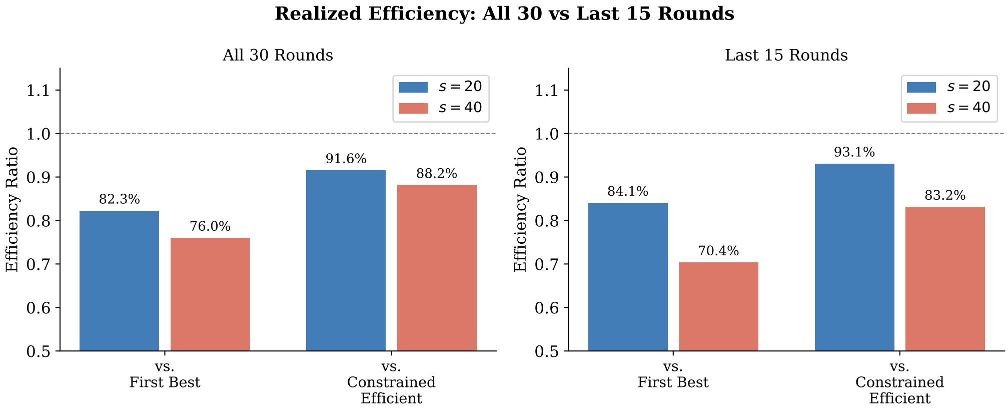


Efficiency — All 30 Rounds



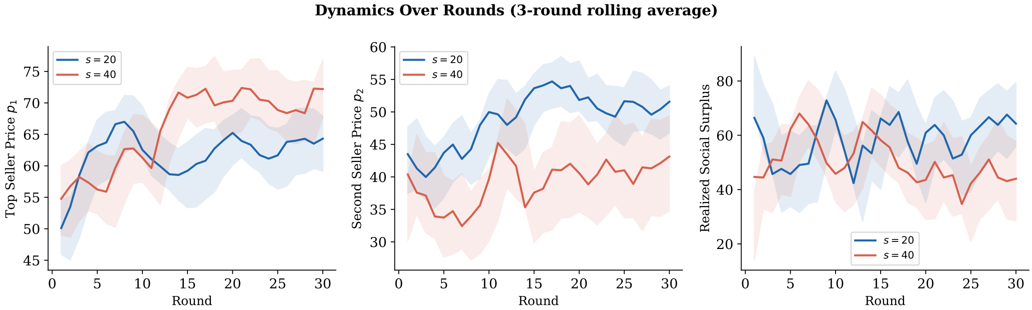
The market achieves 82% of First Best under low prominence and 76% under high prominence. Relative to the Constrained Efficient benchmark, efficiency is 92% and 88% respectively.

Efficiency — Last 15 Rounds



Under $s = 20$, efficiency is roughly stable. Under $s = 40$, efficiency vs. First Best falls below 70% in later rounds — a meaningful deterioration driven by rising top seller prices and falling buyer surplus.

Dynamics Over Rounds



The divergence across treatments is visible in the trajectories: top seller prices rise and realized surplus falls under $s = 40$, while the picture is more stable under $s = 20$.

Summary

- **Price ordering confirmed**: prominent firm prices stochastically higher — prominence confers market power, not competitive pressure.
- **Overbidding**: subjects bid $\sim 4\times$ the equilibrium prediction, especially early on.
- **Over-search**: buyers search more than theory predicts when $V_1 = 0$; they learn to search less over time.
- **Learning diverges**: low prominence converges toward rationality; high prominence becomes more distorted with experience.
- **Welfare cost of prominence**: efficiency falls from 82% to 76% of First Best; DWL rises and is amplified by learning under high prominence.

Broader Implications and Next Steps

- **Platform design**: raising search friction extracts modest revenue but generates substantial consumer harm — and the damage grows as sellers learn to exploit their position.
- **Price observability**: hidden prices would restore the Diamond Paradox — a natural extension for comparison.
- **Behavioral modeling**: estimate quantal-response search; test backward-induction deviations in bidding.
- **Further treatments**: random assignment vs. second-price auction; $N > 2$ firms; quality heterogeneity.

Limitations and Extensions

- **Sample size**: six sessions (3 per treatment) limit statistical power; results are suggestive but underpowered for fine-grained heterogeneity analysis.
- **Symmetric valuations**: the model assumes $V_j \in \{V, 0\}$ i.i.d.; richer value distributions may alter the shape of the MSNE CDFs and fire-sale predictions.
- **Exogenous search order**: prominence is modeled as fixed top-position visibility; endogenous search order with advertising intensity is a natural generalization.
- **Repeated-game effects**: subjects interact within fixed sessions; supergame motives may partly explain persistent overbidding and pricing above equilibrium supports.
- **Platform revenue design**: optimal auction format (e.g., reserve prices, VCG) is left open — the current second-price setup may not maximize platform surplus.

Thank You

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Questions and comments welcome.

Appendix: Why Seller Profits Are Not Zero

A common misconception: since the winner pays the second bid, profits are fully dissipated.

The auction dissipates only the *incremental* value of the top position:

$$\begin{aligned} \text{Win top slot:} \quad & \bar{\Pi}_1 - b^{(2)} = Vq(1-q) - (1-q)s = (1-q)(qV - s) = \bar{\Pi}_2 \\ \text{Lose:} \quad & \bar{\Pi}_2 = \left(V - \frac{s}{q}\right) q(1-q) \end{aligned}$$

Both yield the same net profit $\bar{\Pi}_2 > 0$. For q=0.5, V=100: $\bar{\Pi}_2 = 15$ at s=20; $\bar{\Pi}_2 = 5$ at s=40.

Buyers also extract positive surplus because prices mix below V: a buyer with $V_1 = V$ pays $p_1 < V$ and keeps $V - p_1 > 0$.

Appendix: MSNE — Summary of Derivation

1. **Upper bounds**: surplus-extraction pins $\bar{p}_1 = V$ and $\bar{p}_2 = V - s/q$.
2. **Equilibrium profits** anchored at upper bounds: $\bar{\Pi}_j = \bar{p}_j \cdot q(1-q)$.
3. **CDFs** from mutual indifference: F_1 from Firm 2's indifference; F_2 from Firm 1's.
4. **Mass point**: the continuous candidate $\tilde{F}_2$ has a lower bound where Firm 2 cannot reach $\bar{\Pi}_2$ even at maximum demand q. That probability mass is compressed into an atom at $\underline{p}_2 = (1-q)(V - s/q)$.
5. **Uniqueness**: any gap or interior mass in F_j unravels the indifference condition.

Appendix: Demand Functions

Case	Condition	Demand
Firm 1 (prominent, seen first for free)		
$V_1 = V$, no search	$p_1 - p_2 < s/q$	q
$V_1 = V$, search	$p_1 - p_2 \geq s/q$	q^2
$V_1 = 0$, no search	$p_2 > V - s/q$	0
$V_1 = 0$, search	$p_2 \leq V - s/q$	$q(1-q)$
Firm 2 (non-prominent, costs s to inspect)		
$V_1 = V$, searched	$p_1 - p_2 \geq s/q$	$q(1-q)$
$V_1 = 0$, searched	$p_2 \leq V - s/q$	q^2

With $q = 0.5$: unsearched demand for Firm 1 is 0.5; searched demand falls to 0.25 (buyer prefers 2 if $V_2 = V$, $V_1 = V$).

Appendix: MSNE Equilibrium Conditions

Firms mix over prices so that each is indifferent across its support. Let F_j be Firm j's CDF.

Firm 1's profit at price p_1:

$$\Pi_1(p_1) = p_1\big[q(1-q)(1-F_2(p_1-s/q)) + q(1-q)F_2(p_1-s/q)\cdot q + \ldots\big]$$

Simplified (dominant demand channel):

$$\Pi_1(p_1) = p_1 \cdot q\big[(1-q) + (1-q)F_2(p_1-s/q)\big] = \bar{\Pi}_1$$

Firm 2's profit at price p_2:

$$\Pi_2(p_2) = p_2 \cdot q(1-q)\big[1-F_1(p_2+s/q)\big] + p_2 \cdot q^2 F_1(p_2+s/q) = \bar{\Pi}_2$$

These two indifference conditions, together with support boundary conditions, uniquely pin down F_1 and F_2.

Appendix: Step 1 — Pricing Upper Bounds

Claim: $\bar{p}_1 = V$ and $\bar{p}_2 = V - s/q$.

Firm 1: A buyer with $V_1 = V$ always buys at $p_1 \leq V$ (no search needed). So $p_1 = V$ is the maximum price at which Firm 1 still sells. Charging above V yields zero demand.

Firm 2: A buyer searches only if $p_2 \leq V - s/q$ (when $V_1 = 0$) or $p_1 - p_2 \geq s/q$ (when $V_1 = V$). The global upper bound on p_2 satisfying both conditions is $V - s/q$: at higher prices Firm 2 can never attract any search.

Equilibrium profits at upper bounds:

$$\bar{\Pi}_1 = V \cdot q(1-q), \qquad \bar{\Pi}_2 = \left(V - \frac{s}{q}\right) q(1-q)$$

These are the payoffs that must hold throughout each firm's mixing support.

Appendix: Step 2 — Anchoring Profits

Because each firm can always deviate to its upper bound and earn $\bar{\Pi}_j$, equilibrium profit is constant at $\bar{\Pi}_j$ over the entire mixing support.

Key implication: both supports must be *connected* intervals (no gaps) and *no interior mass points* for Firm 1, otherwise undercutting would be profitable.

Profit identity used in subsequent steps:

$$\Pi_1(p_1) = \bar{\Pi}_1 \quad \forall\, p_1 \in [\underline{p}_1, V]$$
$$\Pi_2(p_2) = \bar{\Pi}_2 \quad \forall\, p_2 \in [\underline{p}_2, V - s/q]$$

Bid: since winning the auction raises profits from $\bar{\Pi}_2$ to $\bar{\Pi}_1$, the incremental gain is $(1 - q)s$, so $b^* = (1 - q)s$.

Appendix: Step 3 — F_1 from Firm 2's Indifference

Firm 2's profit at price p_2 depends on whether search occurs. Buyers search Firm 2 iff:

- $V_1 = 0$: always search if $p_2 \le V - s/q$ (Firm 2 sets prices in this range)
- $V_1 = V$: search iff $p_1 - p_2 \ge s/q$, i.e., $p_1 \ge p_2 + s/q$

Setting $\Pi_2(p_2) = \bar{\Pi}_2$ and solving for F_1:

$$F_1(p_2 + s/q) = \frac{\bar{\Pi}_2 - p_2 q(1-q)}{p_2 q(1-q)(1/q - 1)}$$

Re-indexing with $p = p_2 + s/q$:

$$\boxed{F_1(p) = 1 - \frac{\bar{\Pi}_1}{p \cdot q(1-q)} \cdot \frac{1}{1 + (1-q)/q}}$$

Support: $p_1 \in [\underline{p}_1, V]$ where $\underline{p}_1 = V - s/q \cdot (1/q - 1 + 1) = V - s(1-q)/q^2 \cdot q = \ldots$ Simplified: $\underline{p}_1 = (V - s/q) + s/q = V - s\frac{1-q}{q}$.

Appendix: Step 4 — F_2 from Firm 1's Indifference

Firm 1's profit at price p_1:

$$\Pi_1(p_1) = p_1 \cdot q\left[(1-q) + q \cdot F_2(p_1 - s/q)\right] = \bar{\Pi}_1$$

(First term: $V_1 = V$ and buyer does *not* search; second: $V_1 = V$, buyer searches but $V_2 = 0$.)

Solving for F_2:

$$\boxed{F_2(p_2) = \frac{1}{q}\left(\frac{\bar{\Pi}_1}{(p_2 + s/q)\cdot q(1-q)} - 1\right)\cdot \frac{1}{1-q}}$$

The *continuous* part of F_2 has lower bound $\tilde{p}_2$ where $F_2(\tilde{p}_2) = 0$:

$$\tilde{p}_2 = \frac{\bar{\Pi}_1}{q(1-q)} - \frac{s}{q} = V - \frac{s}{q}$$

But this means all mass "wants" to pile below $V - s/q$, creating the fire-sale atom.

Appendix: Step 5 — Fire-Sale Mass Point at $\underline{p}_2$

The continuous CDF $\tilde{F}_2$ integrates to less than 1 over $[\underline{p}_2, V - s/q]$. The remaining probability mass is an **atom** at $\underline{p}_2 = (1-q)(V - s/q)$.

Atom size:

$$m_2 = 1 - \int_{\underline{p}_2}^{V-s/q} f_2(p)\, dp = 1 - \frac{q}{1-q}\left(\frac{\bar{\Pi}_1}{\bar{\Pi}_2} - 1\right)^{-1} \ln \frac{V - s/q}{\underline{p}_2}$$

	$s = 20$	$s = 40$
$\underline{p}_2$	30	10
$\bar{p}_2 = V - s/q$	60	20
Atom size m_2	0.57	0.89

At high s nearly all probability is in the atom: Firm 2 commits to a deep "fire sale" price to ensure search occurs.

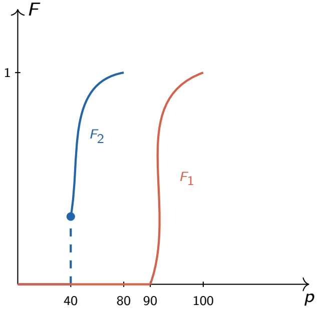
Appendix: Summary of Closed-Form CDFs

	Closed form
$F_1(p_1)$	$1 - \dfrac{\bar{\Pi}_1}{p_1 \cdot q(1-q)} \cdot \dfrac{1}{2-q} \quad \text{on } \left[V - \dfrac{s(1-q)}{q},\ V\right]$
$F_2(p_2)$	$\dfrac{1}{q(1-q)} \left(\dfrac{\bar{\Pi}_1}{p_2 + s/q} - q(1-q) \right) \cdot \dfrac{1}{1-q} \quad \text{on } \left[(1-q)\left(V - \dfrac{s}{q}\right),\ V - \dfrac{s}{q}\right]$
m_2 (atom at $\underline{p}_2$)	$1 - \displaystyle\int_{\underline{p}_2}^{\bar{p}_2} f_2 \, dp$

Both supports are disjoint when $s > 0$: $\bar{p}_2 = V - s/q < V - s(1-q)/q = \underline{p}_1$. The **price gap** between the two supports equals s/q, growing with s.

Appendix: Numerical Example ($s = 10,\ V = 100,\ q = 0.5$)

Quantity	Value
$\bar{p}_1$	100
$\underline{p}_1$	90
$\bar{p}_2$	80
$\underline{p}_2$	40
Atom m_2	0.32
$b^* = (1-q)s$	5
$\bar{\Pi}_1$	25
$\bar{\Pi}_2 = \Pi^{\text{net}}$	20



Appendix: Truthful Bidding in the Second-Price Auction

Theory: each firm's dominant strategy is to bid its incremental value of the top slot:

$$b^* = \bar{\Pi}_1 - \bar{\Pi}_2 = (1 - q)s$$

Observed vs. theory (winning bids, all 30 rounds):

	$s = 20$	$s = 40$
Theory b^*	10	20
Observed mean (win)	44.7	53.8
Observed mean (lose)	22.1	28.3
Ratio (win/theory)	4.5×	2.7×

Persistent overbidding is consistent with subjects anchoring on V rather than the incremental value. Over time winning bids fall (toward 37.9 and 51.3 in the last 15 rounds), suggesting partial learning.

Appendix: Price Gap Widens with Prominence Friction

Theory predicts the two firms' pricing supports are *disjoint*:

$$\bar{p}_2 = V - \frac{s}{q} < V - \frac{s(1-q)}{q} = \underline{p}_1$$

The gap between the supports equals $\frac{s}{q} - \frac{s(1-q)}{q} = s$.

Empirically, the mean price gap $\bar{p}_1 - \bar{p}_2$ is:

	$s = 20$	$s = 40$
Observed mean gap	12.9	27.0
Theory gap (support separation)	20	40
Last 15 rounds	14.5	31.2

The gap roughly doubles with s (qualitatively confirmed). The magnitude is compressed relative to theory, consistent with above-theory pricing by Firm 2 and below-theory pricing by Firm 1.

Appendix: Relation to Literature

Paper	Setup	Key difference
Armstrong, Vickers & Zhou (2009)	Exogenous prominence; consumers search optimally	Prominence given, not auctioned; no endogenous firm value of position
Armstrong & Zhou (2011)	Endogenous prominence via advertising	Ad cost is sunk; here cost is the second bid (a transfer to the platform)
Haan & Moraga-González (2011)	Position advertising with directed search	Symmetric firms; no mixed-strategy price equilibrium with fire-sale atom
This paper	Auction for prominence + directed search	Endogenous valuation of position; second-price auction; fire-sale atom; lab experiment